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Can Quantum Computers Break the Internet?

The Technology That Could Change Cybersecurity

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Founder, ThinkHub



Can quantum computers break the internet?



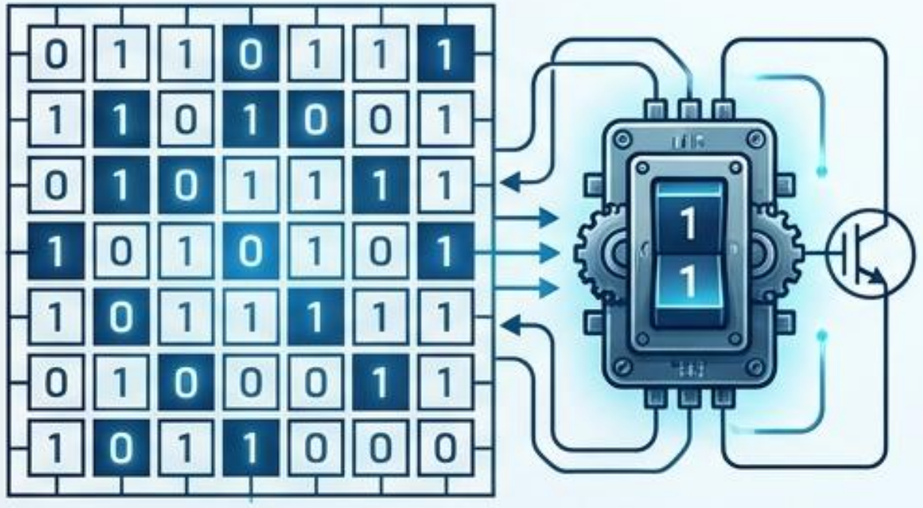
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TWO DIFFERENT RULES OF COMPUTATION

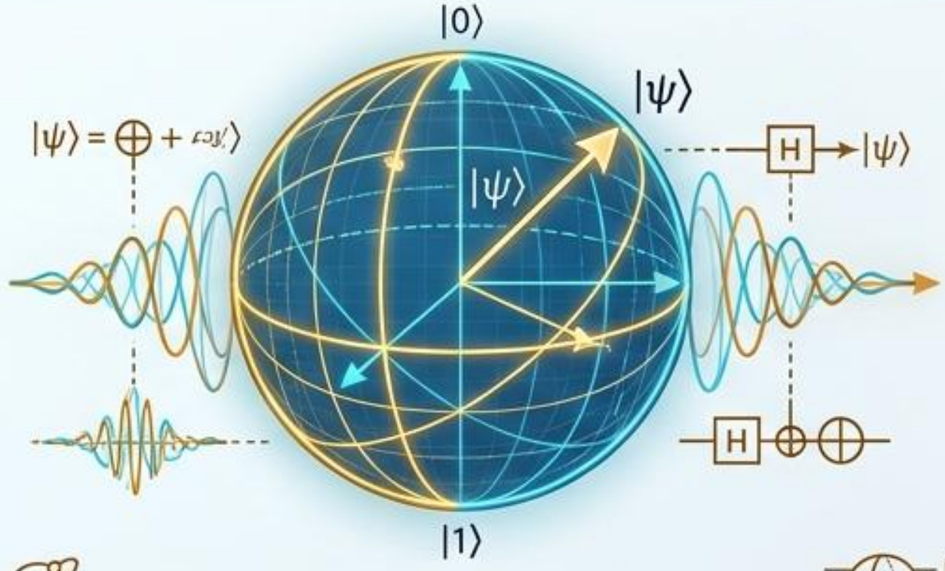
CLASSICAL COMPUTING



- **Information Unit:** Bits (0 or 1).
- **State Manipulation:** Classical logic gates.
- **Computation:** Linear execution of classical algorithms.
- **Best For:** Broad, general-purpose workloads (web servers, smartphones, AI processing).

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QUANTUM COMPUTING



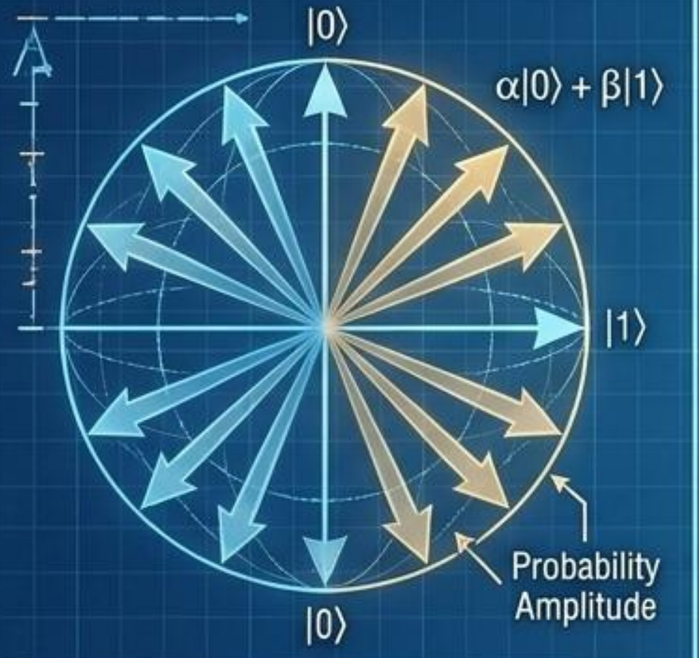
- **Information Unit:** Qubits (quantum states).
- **State Manipulation:** Controlled quantum operations.
- **Computation:** Manipulation of probability amplitudes via interference.
- **Best For:** Specialized problems with defined mathematical structures.

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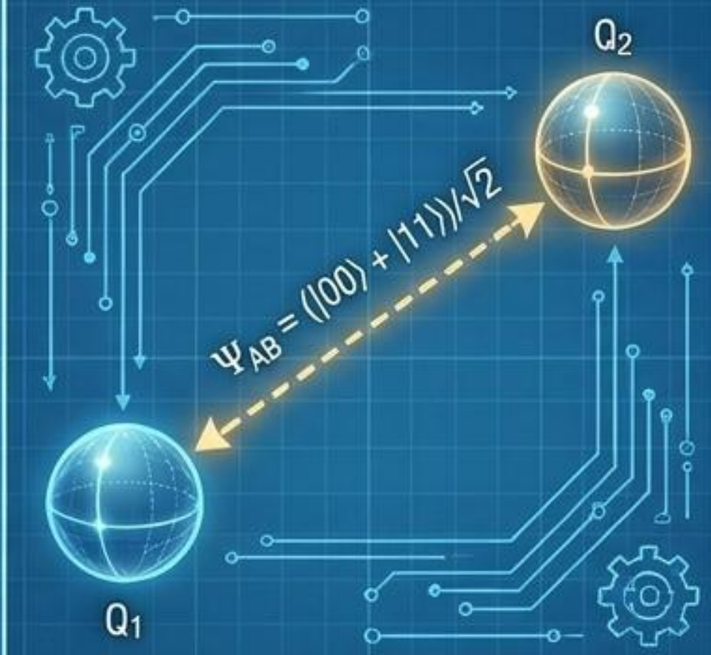
DECODING THE QUANTUM STATE

SUPERPOSITION



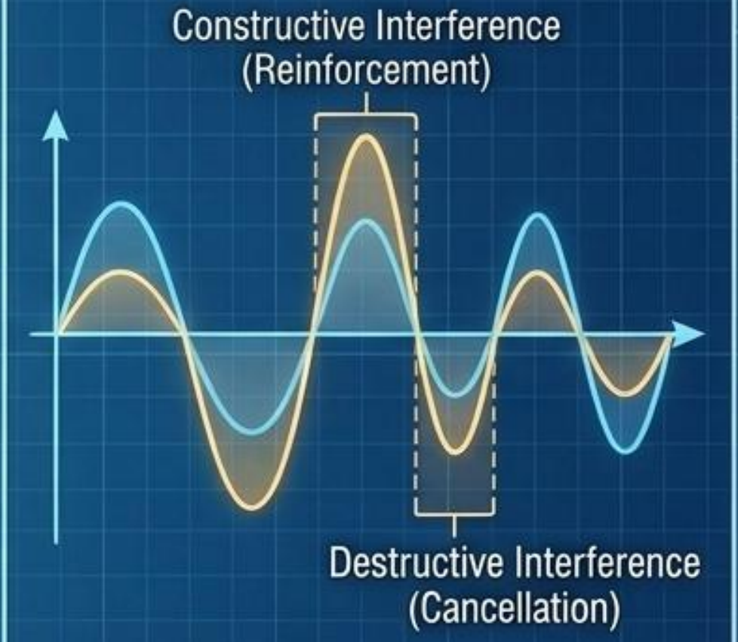
A qubit holds a state involving multiple probability amplitudes before measurement. It is not simply "0 and 1 at the same time." Measuring it collapses it to a single outcome.

ENTANGLEMENT



Multiple qubits share a joint state. Measuring one instantly correlates with the other. It is a mathematical link, not a faster-than-light communication channel.

INTERFERENCE



The true engine of quantum algorithms. Carefully designed operations cause the correct answers to reinforce each other, while incorrect answers cancel out.

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INSIDE A QUANTUM ARCHITECTURE



1. **CLASSICAL HOST:** User writes a program and selects a quantum algorithm.
2. **COMPILER:** Translates the algorithm into hardware-compatible quantum gates.
3. **CONTROL SYSTEM:** Microwave or laser pulses manipulate the physical hardware.
4. **QUANTUM PROCESSOR:** Physical qubits execute state changes inside an extreme cooling environment.
5. **MEASUREMENT:** The delicate quantum state is measured and collapsed into classical data.
6. **POST-PROCESSING:** Classical software analyzes the statistical outcomes.

KEY TAKEAWAY

A quantum processor does not work alone. It requires massive classical infrastructure, extreme chilling, and precise environmental control.

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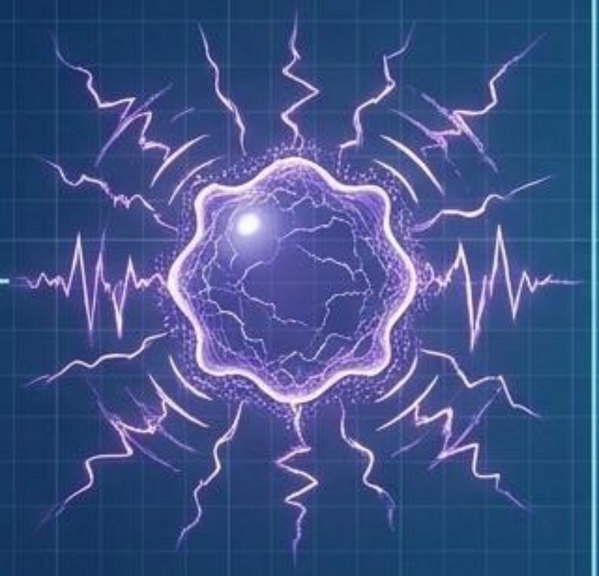
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THE RELIABILITY PROBLEM: PHYSICAL VS. LOGICAL QUBITS

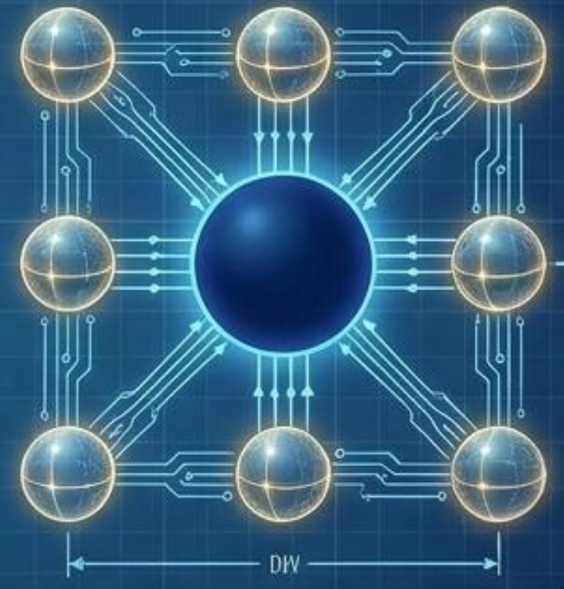
Quantum states are incredibly fragile. Heat, radiation, and even measurement can introduce fatal errors into a calculation.

Physical Qubits:
The raw hardware units. When headlines boast about 'high qubit counts," they mean inherently noisy physical qubits.

PHYSICAL QUBITS



LOGICAL QUBITS



Logical Qubit:
An error-corrected unit created by grouping many physical qubits together.

TAKEAWAY: Breaking real-world cryptography requires thousands of perfectly stable logical qubits, demanding a massive overhead of physical qubits that does not yet exist.

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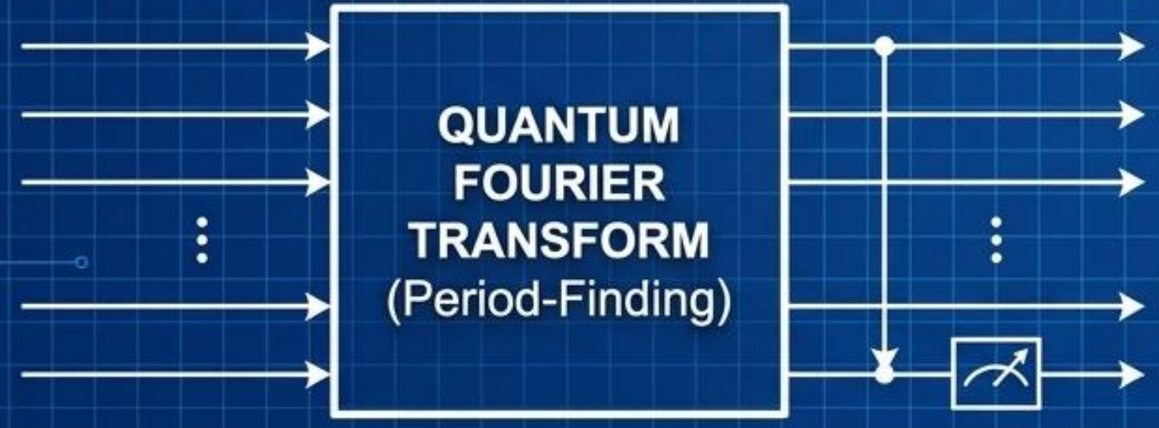
SHOR'S ALGORITHM: THE MATHEMATICAL THREAT

Modern public-key encryption (like RSA) relies on a simple premise:

$17 \times 19 \rightarrow 323$ (Easy to Compute)

$323 \rightarrow ?$ (Practically Impossible for Classical Computers)

HOW QUANTUM BREAKS IT



Introduced in 1994, Shor's Algorithm doesn't just guess factors millions of times faster. Instead, it uses quantum interference to map the periodic structure of the math. It mathematically cancels out the wrong answers and amplifies correct prime factors, efficiently solving the integer-factorization problems that secure the internet.

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How HTTPS Protects Your Data (And Where It's Vulnerable)



1. Negotiate Connection

Browser and server agree on TLS parameters.

[SAFE]

2. Validate Identity

Server presents a certificate. (Uses Public-Key Crypto)

[VULNERABLE]

3. Establish Shared Key

Parties mathematically agree on a secret key over the open internet. (Uses Public-Key Crypto)

[VULNERABLE]

4. Protect Data

Bulk data (passwords, bank balances) is locked with symmetric encryption. (Uses AES)

[SAFE]



HTTPS isn't a single algorithm; it's a stack. A quantum computer wouldn't break the whole stack, but by defeating the key establishment phase, the rest of the security is compromised.

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What Quantum Computers Break (And What They Don't)

Vulnerable - Requires Replacement



- **RSA Encryption:** Underlying math is broken by Shor's algorithm.



- **Elliptic Curve (ECC):** Vulnerable to quantum discrete-logarithm algorithms.



- **Diffie-Hellman:** Key exchange protocols face structural collapse.

Resilient - Requires Larger Keys/Analysis



- **AES (Symmetric Encryption):** Unaffected by Shor's algorithm. Grover's algorithm only provides a speedup that can be mitigated simply by using larger key sizes (e.g., AES-256).



- **Hash Functions (SHA):** Largely secure; quantum algorithms only change the analysis of collision resistance.



- **Passwords:** Not automatically defeated; rely on hashing and robust authentication, not public-key factoring.

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The data may outlive the encryption protecting it.



Capture
(Present Day)

Store

Capability
(Future)

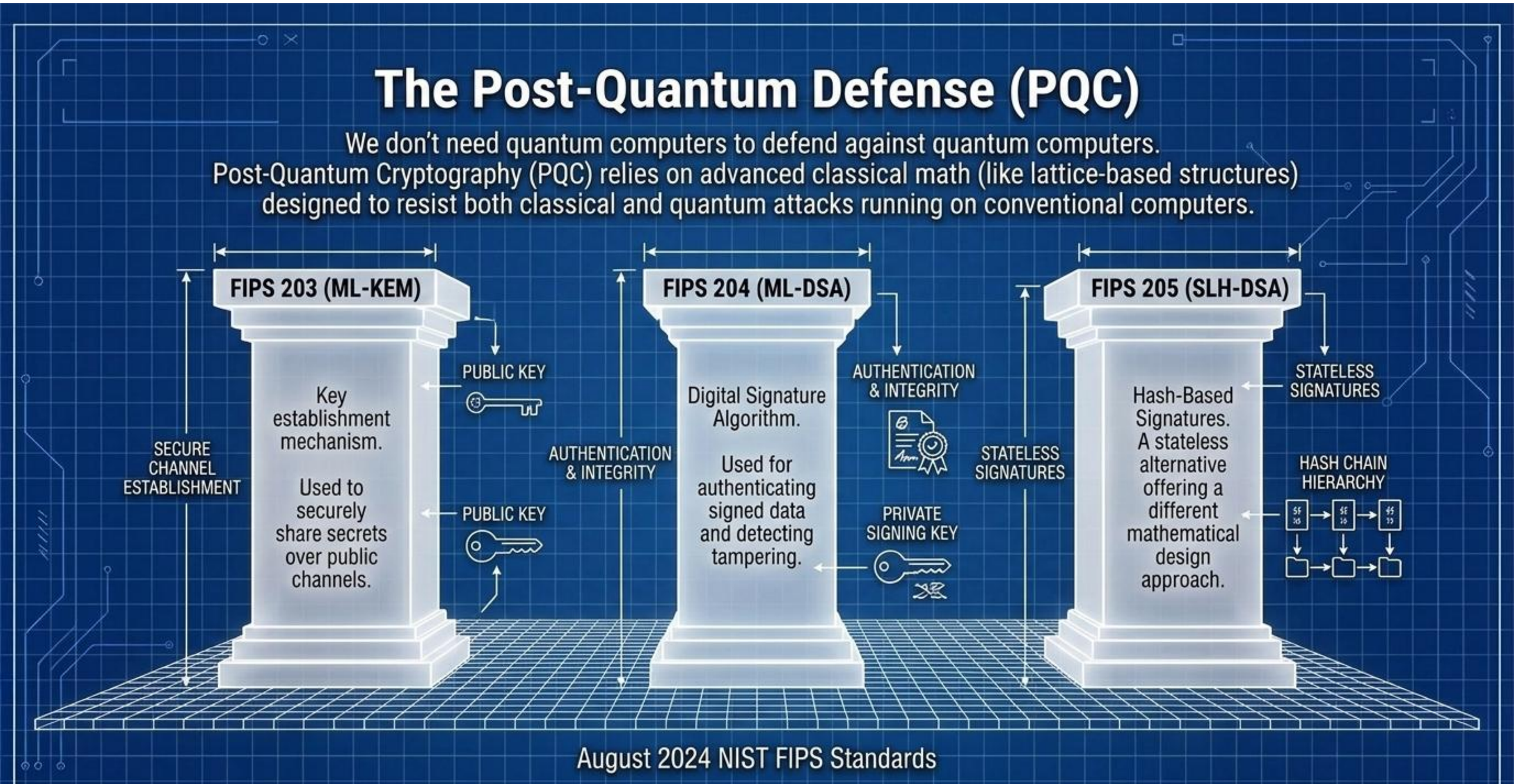
Exposure

Why urgency matters now: If your data must remain secret for 15+ years (government intelligence, health records, trade secrets), it is already vulnerable today.

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The Post-Quantum Defense (PQC)

We don't need quantum computers to defend against quantum computers.
Post-Quantum Cryptography (PQC) relies on advanced classical math (like lattice-based structures) designed to resist both classical and quantum attacks running on conventional computers.



August 2024 NIST FIPS Standards

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The Internet's Cryptographic Upgrade

1. **Discover:** Inventory dependencies (embedded devices, certificates, software).
2. **Prioritize:** Identify long-lifespan data at risk of "Harvest Now" attacks.
3. **Test:** Evaluate memory, latency, and interoperability of new algorithms.
4. **Deploy:** Execute controlled rollouts via standard internet protocols.
5. **Monitor:** Maintain cryptographic agility to swap algorithms if flaws are found.

The Transition Strategy: Hybrid Key Exchange (RFC 9954 / 10024)

By combining traditional (e.g., Elliptic Curve) and post-quantum (ML-KEM) key exchanges in TLS 1.3, connections remain secure as long as at least one of the algorithms remains unbroken.



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India's National Quantum Mission (NQM)

Timeline:
2023—2031
(Approved April 2023)

Outlay:
₹6,003.65 Crore

Target:
50 to 1,000 physical qubits over eight years.

Computing
Developing intermediate-scale quantum hardware.

Communication
Satellite-based secure networks and inter-city key distribution.

Sensing
Advanced metrology and precision devices.

Enabling Technologies
Building the surrounding engineering ecosystem.

Research vs. Reality: These are strategic capacity-building targets. They signify a massive domestic investment in quantum ecosystems, but do not mean India has currently deployed a machine capable of breaking modern encryption.

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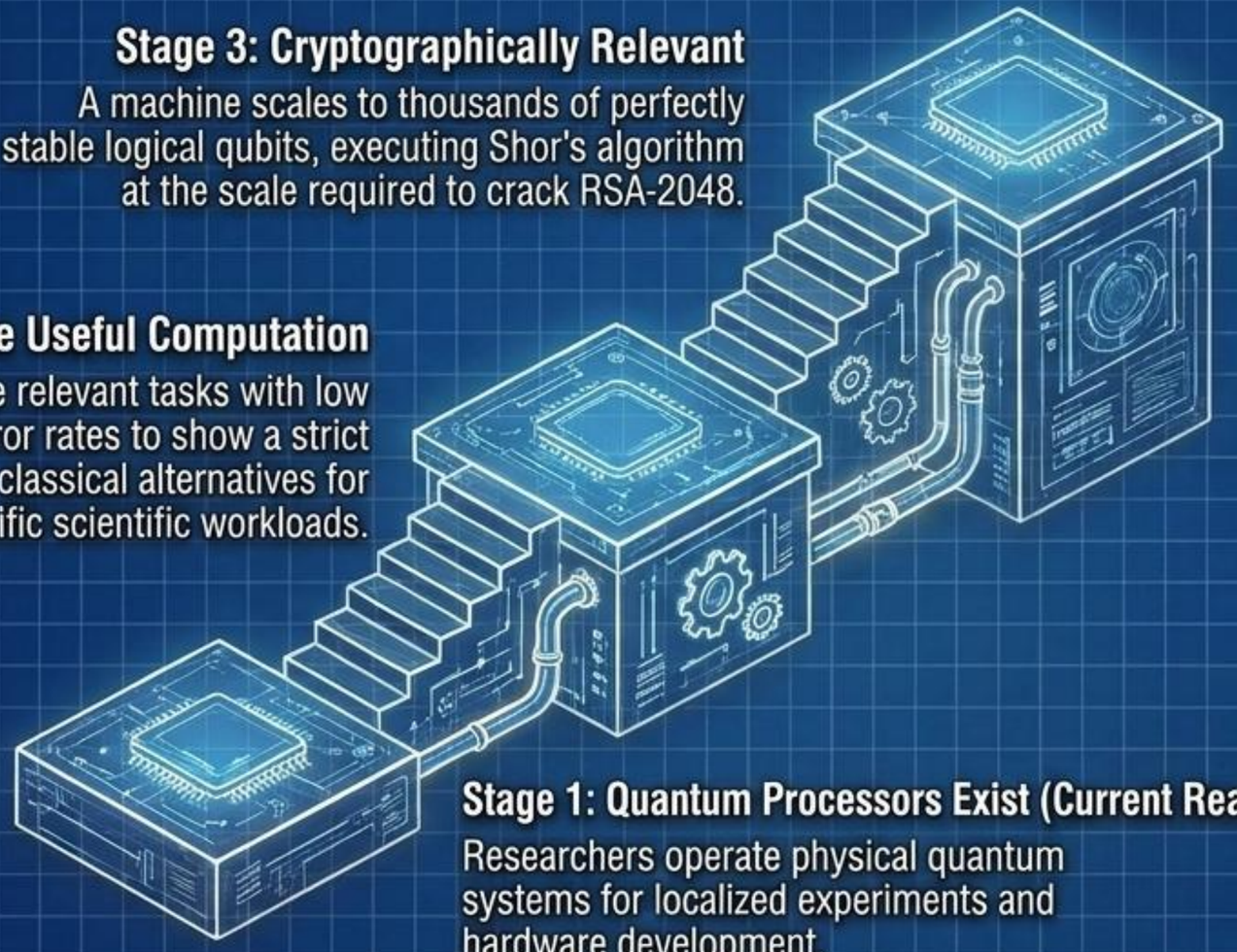
How Far Away Is the Quantum Threat?

Stage 3: Cryptographically Relevant

A machine scales to thousands of perfectly stable logical qubits, executing Shor's algorithm at the scale required to crack RSA-2048.

Stage 2: Reliable Useful Computation

Systems execute relevant tasks with low enough error rates to show a strict advantage over classical alternatives for specific scientific workloads.



Stage 1: Quantum Processors Exist (Current Reality)

Researchers operate physical quantum systems for localized experiments and hardware development.

There is no universally accepted arrival date. Corporate roadmaps (like IBM's target of 200 logical qubits by 2029) are engineering goals, not guaranteed dates for cryptographic doom. However, because infrastructure migration takes years, preparation must begin under uncertainty.

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What Quantum Computers Cannot Magically Do

Myth	Reality
They test all possible answers simultaneously.	**They use interference to manipulate probabilities,** amplifying correct answers and canceling wrong ones.
Entanglement allows faster-than-light communication.	**It creates correlated states, but cannot be used to transmit usable data faster than light.**
A quantum machine will break the entire internet overnight.	**They only threaten specific public-key math.** Passwords, hashes, and symmetric data encryption (AES) remain largely secure.
We must buy quantum hardware to stay safe.	**Post-Quantum Cryptography** (PQC) standards are classical algorithms designed to run on the standard computers, servers, and smartphones we use today.

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The quantum era is a reason to prepare—not panic.



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THANK YOU!

The future belongs to those who build it continuously.

START TODAY.
Take the first step.

STAY CONSISTENT.
Small steps, big results.

BUILD ONE PROJECT
at a time.



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